# SCD Assignment #2
# Web Scraping Using Beautiful Soup

## Instructions:

1. Write your Python code using Beautiful Soup and Requests libraries.
2. Provide appropriate comments in your code to explain your approach.
3. For each task, show the output.
4. All tasks carry equal marks.
5. Your final submission should include a PDF file containing all the source code and their outputs, Python Code file(s) and XML files.
6. Half the marks will be deducted for plagiarism from a fellow student.

## Overview:

Epic Games Store is a video game digital distribution service and storefront operated by Epic Games. In this assignment, your task is to first meticulously inspect the HTML structure of the webpage then design a web scraper that scrapes the webpage to extract the relevant details for each game, process the data, and export it into an XML file. The URL for the webpage is given below:
URL: https://store.epicgames.com/

## Tasks:

1. Write a Python script that retrieves the HTML card/div structure of any one of the listed games and prints its raw HTML content using the **find()** method.

   **Example Output:**

```
<div class="outer-card">
        <div class="box">
                <span class="title"> Dummy 1 </span>
                <span class="price"> $30 </span>
        </div>
</div>
```

2. Write a Python script that only retrieves any two games that are advertised as **"Free".**

   a. Extract the **title** for each game using the **find()** method.

a. Extract the **price** for each game using the **find('tag', {'attribute':'value'})** method.

b. Extract the **link** for each game using the **find('tag', class_='example')** or **find(id='example')** method.

c. Extract the **image source** for each game using the **select_one('.example')** method.

**Example Output:**
****Free Games***:*
*1. Title: Dummy 1*
*- Price: Free*
*- URL: https://store.epicgames.com/en-US/p/dummy1*
*- Image: https://cdn1.epicgames.com/img1*

*2. Title: Dummy 2*
*- Price: Free*
*- URL: https://store.epicgames.com/en-US/p/dummy2*
*- Image: https://cdn1.epicgames.com/img2*

2. Write a Python script that only retrieves any three games that are in the price range of **$5** to **$30**.

   a. Extract the **title** for each game using the **.find_children()** method.

   b. Extract the **price** for each game using the **.find_parent()** method.

   c. Extract the **link** for each game using the **.find_next_sibling()** or **.find_previous_sibling()** method.

   d. Extract the **image source** for each game using the **.descendants** or **.children** property.

**Example Output:**
****Paid Games***:*
*1. Title: Dummy 3*
*- Price: $15*
*- URL: https://store.epicgames.com/en-US/p/dummy3*

*- Image: https://cdn1.epicgames.com/img3*

*2. Title: Dummy 4*
*- Price: $10*
*- URL: https://store.epicgames.com/en-US/p/dummy4*
*- Image: https://cdn1.epicgames.com/img4*

*3. Title: Dummy 5*
*- Price: $29*
*- URL: https://store.epicgames.com/en-US/p/dummy5*
*- Image: https://cdn1.epicgames.com/img5*

1. Generate an XML file with all the scraped data, organizing each game as an individual XML element with child elements for title, price, URL and image source.

**Example XML File:**

<?xml version='1.0' encoding='utf-8'?>
<games>
    <game>
        <title>Dummy 1</title>
        <price>Free</price>
        <url>https://store.epicgames.com/en-US/p/dummy1</url>
        <img>https://cdn1.epicgames.com/img1</img>
    </game>
    <game>
        <title>Dummy 2</title>
        <price>Free</price>
        <url>https://store.epicgames.com/en-US/p/dummy2</url>
        <img>https://cdn1.epicgames.com/img2</img>
    </game>
    <game>
        <title>Dummy 3</title>
        <price>$15</price>
        <url>https://store.epicgames.com/en-US/p/dummy3</url>
        <img>https://cdn1.epicgames.com/img3</img>
    </game>
    <game>
        <title>Dummy 4</title>

*<price>$10</price>*
*<url>https://store.epicgames.com/en-US/p/dummy4</url>*
*<img>https://cdn1.epicgames.com/img4</img>*
*</game>*
*<game>*
*<title>Dummy 5</title>*
*<price>$29</price>*
*<url>https://store.epicgames.com/en-US/p/dummy5</url>*
*<img>https://cdn1.epicgames.com/img5</img>*
*</game>*
*</games>*

**Important Note:**

In case of Error: 403 while requesting the webpage, use the following headers with the request object:

```
url = "https://store.epicgames.com/"

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124
Safari/537.36",
    "Referer": "https://store.epicgames.com/",
    "Accept-Language": "en-US,en;q=0.9"
}

response = requests.get(url, headers=headers)
```

- *Also understand role of User Agents*